PAPER_1572_CONCRETE_DENSITY_2400

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PAPER_1572 — Concrete Density = ρ_concrete = SO_5²·D_phys·D_BSFG = 100·24 = 2400 kg/m³

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Engineering
Date: June 18, 2026
Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S567 (Engineering Unified Proof Set) lists **Concrete Density** as an EXACT-tier closure:

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```
ρ_concrete = SO_5²·D_phys·D_BSFG = 100·24 = 2400 kg/m³
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UQFF Closed Identity

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```
ρ_concrete = SO_5²·D_phys·D_BSFG = 100·24 = 2400 kg/m³ EXACT
```

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Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Y_S567
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "concrete_density_2400"})`
